// FROM PROBLEM SOLVING TO AUTONOMOUS COGNITIVE GUIDANCE

AGENTIC AI IN MATH EDUCATION

> *The math AI category is moving from answer generation toward autonomous, pedagogically guided reasoning support.*

U.S. MARKET 2026

$2.73B

2035 PROJECTION

$39.83B

// STRATEGIC_TRANSFORMATION

MATH IS THE PROVING GROUND

> *Math reveals whether AI is creating answers or learning*

RCT STUDENTS	STRATEGIC FOCUS
839	Reasoning

STRATEGIC_EXECUTION_MATRIX

01 Mathematics requires sequential reasoning, misconception diagnosis, hinting, feedback, and transfer.

02 These are exactly the capabilities that distinguish agentic cognitive guidance from generic problem solving.

03 The category opportunity is not faster answers; it is better reasoning guidance.

// STRATEGIC_TRANSFORMATION

2026 IS THE MARKET INFLECTION POINT

> *AI education is entering a high-growth adoption cycle*

2026 MARKET	CAGR
$2.73B	*34.8%*

STRATEGIC_EXECUTION_MATRIX

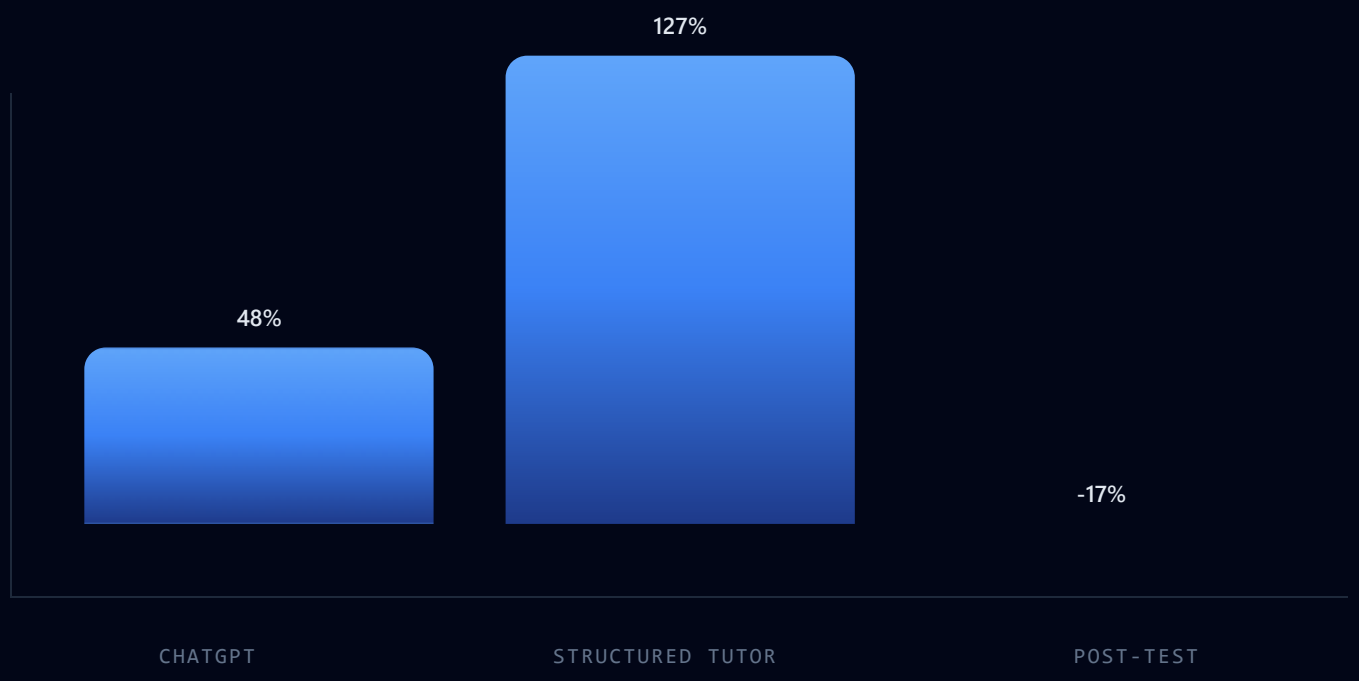
01 Personalized learning, adaptive platforms, and tutoring are accelerating AI adoption.

02 Analytics and automated assessment are critical drivers of this expansion.

03 Agentic math tutoring sits inside a rapidly expanding AI education market.

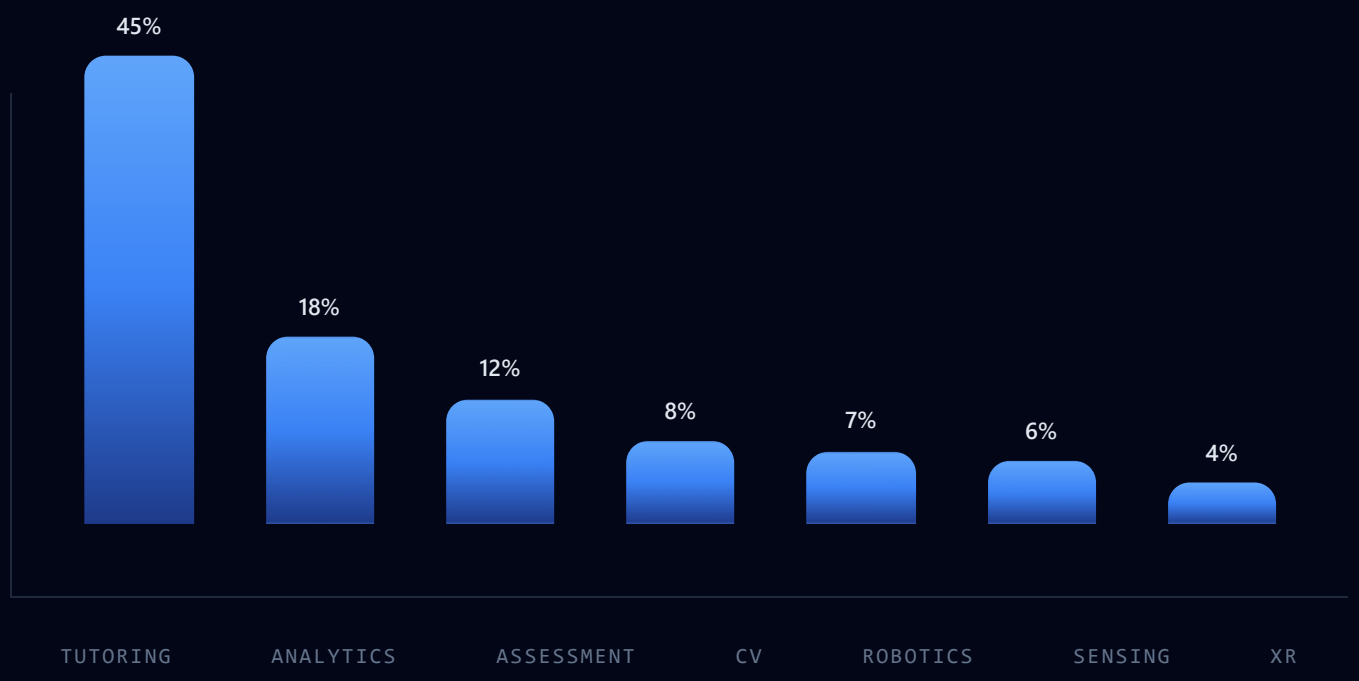
EVIDENCE: DESIGN DETERMINES LEARNING

Unguided AI can become a crutch



RESEARCH MOMENTUM FAVORS TUTORING

Studies cluster around agentic guidance foundations



// STRATEGIC_TRANSFORMATION

THE WINNING STACK

> *Aligning with current market structure*

SOLUTIONS	ML SHARE
70.3%	*64.7%*

STRATEGIC_EXECUTION_MATRIX

01 Solutions revenue share leads at 70.3%.

02 Machine learning dominates application technology at 64.7%.

03 Cloud deployment remains the standard at 60.1%.

04 Learning platforms and virtual facilitators share reached 45.9%.

// STRATEGIC_TRANSFORMATION

TEACHERS ADOPT AI — TRUST IS UNRESOLVED

> *Adoption and integrity concerns rise together*

TEACHER USE	INTEGRITY RISK
37%	*72%*

STRATEGIC_EXECUTION_MATRIX

01 37% of lower-secondary teachers are already using AI for work.

02 57% of teachers say AI helps with lesson planning.

03 72% are concerned AI can harm academic integrity.

04 Teacher trust is the gating factor for scaled classroom adoption.

FROM SOLVER TO AUTONOMOUS COGNITIVE GUIDE

- >> Calculator / Step solver: Traditional tools with low pedagogical autonomy.
- >> Chatbot: Conversational but often generic in problem solving.
- >> AI Tutor: Uses scaffolding and reflection for better learning.
- >> Agentic Cognitive Guide: Planning and memory for autonomous coaching.

ITS APPLICATION

45.9%

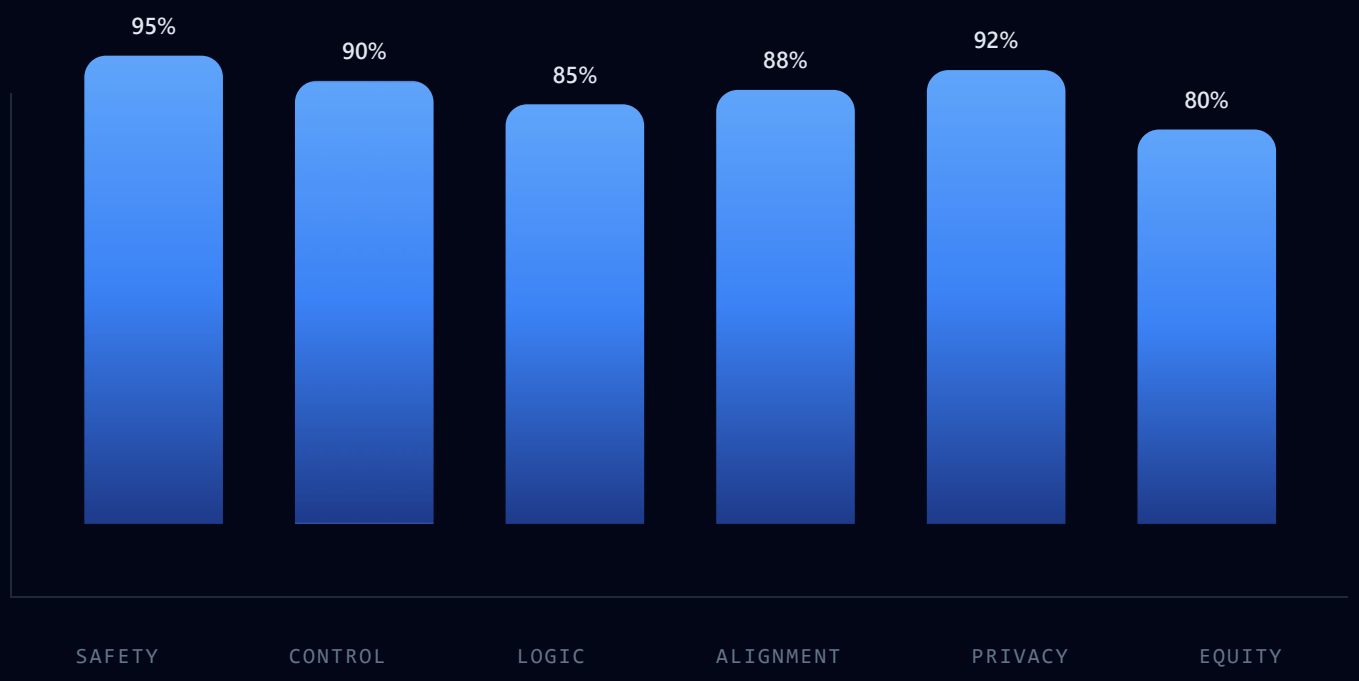
TUTOR LIFT

127%

STRATEGIC SUMMARY COMPLETE

SCHOOL PROCUREMENT REQUIREMENTS

What schools require before scaling



"GenAI can support learning when guided by clear teaching principles. However, if used without guidance, outsourcing tasks to GenAI simply enhances performance with no real learning gains."

OECD REPORT

2026 OUTLOOK // STRATEGIC

EXECUTIVE RECOMMENDATION

- >> Prioritize Durable Learning: Target the +127% performance lift.
- >> Build Teacher Trust: Address the 72% integrity concern directly.
- >> Scale Personalization: Leverage the 34.8% market CAGR.
- >> Guided Autonomy: Ensure AI is constrained by pedagogy and teacher agency.

2026 MARKET

$2.73B

2035 GOAL

$39.83B

STRATEGIC SUMMARY COMPLETE